

Yuan-Jyun Lee

Taipei, Taiwan

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RESEARCH INTERESTS

Operator Algebras, Non-commutative Dynamical Systems, Quantum Markov semi-groups, Ergodic Properties

EDUCATION

National Yang Ming Chiao Tung University (NYCU)

B.S. in Department of Applied Mathematics

Cumulative GPA: 4.01/4.30

Taiwan

Fall 2022 — Present

National Yang Ming Chiao Tung University (NYCU)

B.S. in Department of Electronics and Electrical Engineering

Taiwan

Fall 2023 — Present

HONORS

Academic Excellence Awards

Awarded for outstanding academic performance

Hsinchu, Taiwan

Spring 2022, Fall 2023

College of Science Dean Award

For ranking in the top 10% of the department in my first three years of undergraduate study.

Hsinchu, Taiwan

2023

Zhu Shun-Yi Cooperative Scholarship

Scholarship for academic excellence and contribution to research

Hsinchu, Taiwan

2025

Mathematics Presidential Award

Awarded by NYCU Applied Mathematics Department for outstanding performance

Hsinchu, Taiwan

Spring 2022, Fall 2022, Fall 2023

Excellence in Undergraduate Research

Awarded for outstanding results in undergraduate research projects

Hsinchu, Taiwan

2025

PROJECTS

Research reading group on QMS theory

Hsinchu, Taiwan

Independent research in QMS theory

Spring 2025 — Present

Advisor: Cheng-Fang Su

- Studying mathematical foundations of quantum Markov semigroups and open quantum system dynamics.
- Reviewing topics including completely positive trace-preserving (CPTP) maps, GKSL theorem, Fregiro's theorem and functional calculus for generators.
- Investigating connections between Markovian and non-Markovian quantum dynamics, including process tensor formulations.

Research reading group on ergodic theory

Hsinchu, Taiwan

Studied foundational results in ergodic theory

Summer 2025

Advisor: Xiang Fang

- Studied fundamental topics in ergodic theory, including measure-preserving transformations and ergodic theorems.
- Prepared detailed lecture notes and expository materials for instructional use.
- Primary reference: Walters, *An Introduction to Ergodic Theory* (2000).

Undergraduate Summer Research Program, TWSIAM

Hsinchu, Taiwan

A Study on Trace Inequalities Derived from Relative Entropy

Fall 2024 — Spring 2025

Advisor: Cheng-Fang Su, Hao-Wei Huang

- Studied mathematical foundations of quantum information theory, focusing on entropy inequalities.
- Established preliminary trace inequality results under structural constraints on density matrices, contributing to ongoing research in the area.

Undergraduate Summer Research Program, TWSIAM

Hsinchu, Taiwan

A research on finite time Kuramoto model

Advisor: Ming-Cheng Shiue

Summer 2023

- Studied synchronization phenomena in the Kuramoto model.
- Improved bounds on convergence rates for finite-time variants of the model.

CONFERENCE

Oral Presentation, *Finite time convergence of Kuramoto model*

2024 NTNU Fall Applied Mathematics Student Conference

Recipient of Best Presentation Award.

SELECTED COURSES

Graduate Coursework

- Functional Analysis
- Operator Theory II
- Real Analysis I, II(Measure Theory)
- Quantum Mechanics
- Algebraic Graph Theory
- Stochastic Process
- Fluid Dynamics

TEACHING EXPERIENCE

Fall 2025: Calculus I**Fall 2025:** Probability**Fall 2025:** Fundamental Mathematics**OTHER EXPERIENCES**

President, Mathematics Research Club

Organized biweekly research meetings, managed club activities, and facilitated academic discussions

Hsinchu, Taiwan

Fall 2025 – Present

Founder, Department Football (Soccer) Team

Established and organized the NYCU Mathematics Department football (soccer) team

Hsinchu, Taiwan

Fall 2022 – Present

Captain, Department Basketball Team

Led team training and coordinated inter-departmental competitions

Hsinchu, Taiwan

2023

ENGLISH

TOEFL(ibt): 5.5 (100) (overall score)

Listening: 5.5 — Reading: 5.5

Speaking: 3.5 — Writing: 5

Test date: Mar. 2026

SKILLS

- **Programming:** C++, Python, Matlab
- **Languages** Mandarin, English